

Chap 3

1) Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$

Change Domain (row) = $\{1, 2, 3\}$

Range (col) = $\{a, b, c, d\}$

$R = \{(1, a), (1, b), (2, b), (3, a), (3, c), (3, d)\}$

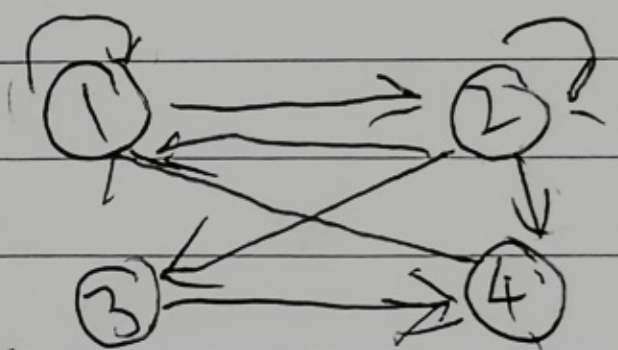
change to matrix =

row \ col	a	b	c	d
1	1	0	0	0
2	0	1	0	0
3	1	0	1	1

2) Digraphs of a relation

Small E: Let $A = \{1, 2, 3, 4\}$ // $R = \{(1, 1), (1, 2), (2, 1), (2, 3), (2, 4), (3, 4), (4, 1)\}$

Draw digraph of the



(col) → Vertex	1	2	3	4
In-degree	2	2	1	2
(row) → Out-degree	2	3	1	1

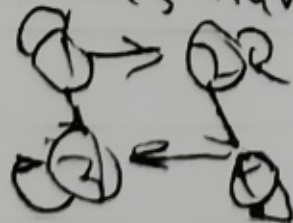
// 自己 - 4 circle In & out 都加一次 (都有算)

to M

3) Properties of relations

- Reflexive - all vertices have loop // 每个圈的值都有自己 loop 才算 reflexive

E:



// R is reflexive

R is not reflexive since $(a, a) \notin R$

- Irreflexive - all vertices have no loop // 每个圈的值都没有自己 loop 才算 irreflexive

E:



// R is irreflexive since $(a, a) \notin R$ (# no repeat inside R)
R is not irreflexive since $(a, a) \in R$

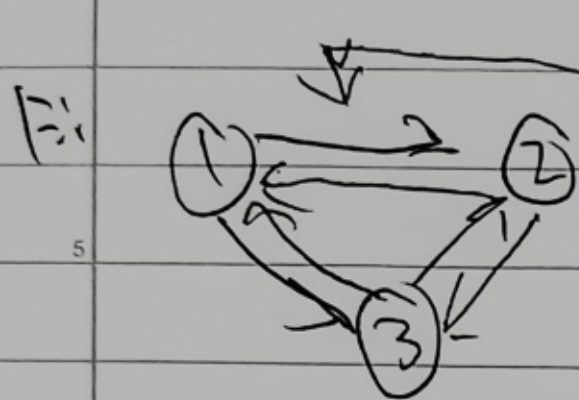
since $(1,2) \in R \notin R$ // means 这个 path 存在 OR 没有 R 里面
 explain is 什么...

Ref:

Date:

• Symmetric (2 ways)

- All the paths are 2 ways // 每个都有来回的



// R is symmetric

R is not symmetric since $(1,2) \in R$ but

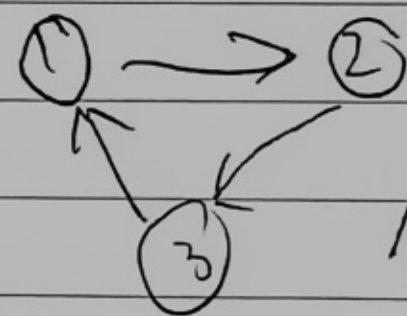


$(2,1) \notin R$

// 1 连 2 在 R 里面, 但 2 连 1 没有

• asymmetric (1 ways, no loop)

- All paths are 1-way, no cycle of length 1 // Ex:



// R is asymmetric

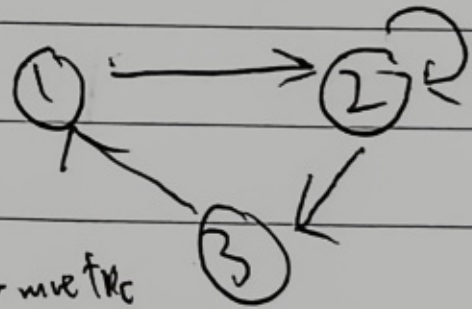
R is not asymmetric since $(a,a) \in R$

有 loop 在 R 里面

• Antisymmetric (1-ways, can have loop)

- loop allowed but must be 1-way path

If got 2-way path, then not antisymmetric



// R is antisymmetric

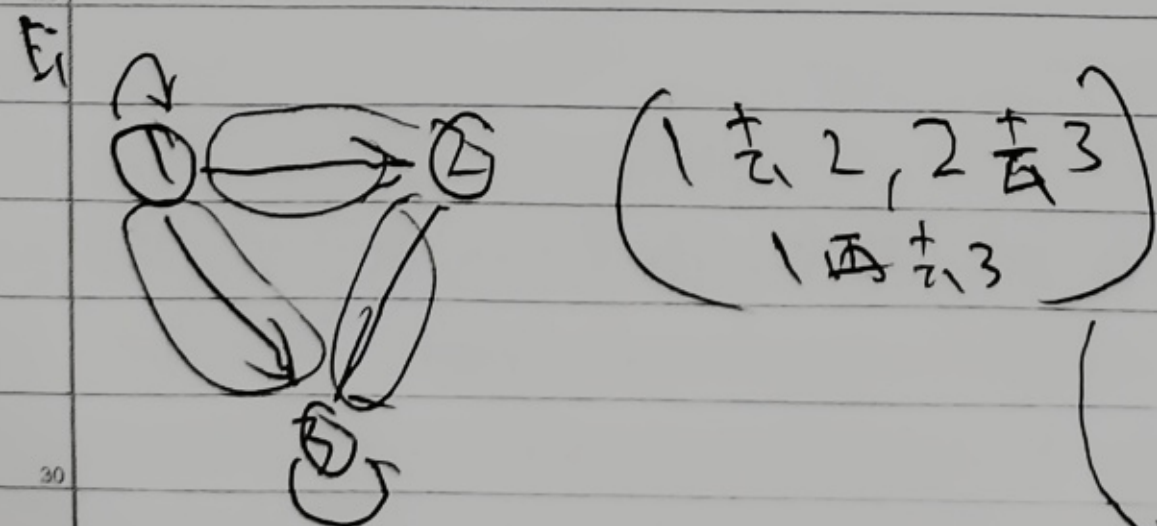
R is not asymmetric since $(a,a) \in R$ and $(b,a) \in R$

有 2-way path, so wrong

Ex: $1 \rightleftharpoons 2$

• Transitive

- R is transitive



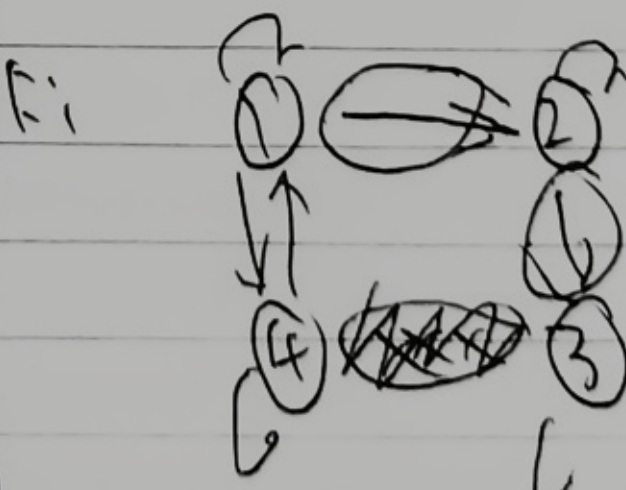
- 需要 $(1,2), (2,3)$

$(1,3)$

- R is not transitive since $(1,2) \in R$

and $(2,3) \in R$

but $(1,3) \notin R$



- 需要
这个条件

// 如果这样就是 not

不只通 logic, 背概念.